

Thomas Xiao

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Education

Carnegie Mellon University

B.S. in Applied Mathematics, Expected May 2030

Pittsburgh, PA

Aug 2026 - Present

- Interests: quantitative research, scientific computing, statistical modeling, and applied artificial intelligence.

Arnold O. Beckman High School

High School Diploma; SAT: 1570 (790 Math, 780 EBRW); weighted GPA: 4.63/5.00

Irvine, CA

Aug 2022 - May 2026

Research Experience

USC Marshall School of Business

Research Assistant - Game Design and Consumer Behavior

Remote

2024 - 2025

- Collaborated with Dr. Cosimo Arnesano to analyze the BoardGameGeek dataset using regression and boosting methods.
- Evaluated associations among ratings, complexity, play time, and consumer response; documented findings for design and marketing decisions.
- Reported findings at CONF-APMM 2025 with an accompanying research video. ResearchGate | Repository

Independent Climate Research Collaboration

Research Collaborator

Remote

2024 - 2025

- Analyzed NOAA U.S. climate records from 1895-2024 using harmonic regression, Fourier analysis, wavelet decomposition, and change-point detection.
- Compared frequency-aware models with trend-only baselines and produced reproducible Python code, figures, and documented results.
- Paper: Detection of Temporal Variability in U.S. Climate Using Harmonic and Wavelet Decomposition.

Experience and Leadership

A2Admission Education

English and Math Tutor

Irvine, CA

2024 - Present

- Provide 4+ hours of individualized K-12 tutoring each week; turn difficult math and writing concepts into clear, step-by-step explanations and adapt instruction to each student.

Beckman Computer Science Club

Board and Program Lead

Irvine, CA

2024 - 2026

- Taught algorithms and data structures to 15+ students and mentored peers in Python and Discord application development.

Beckman MathMedics

Math Instructor

Irvine, CA

Aug 2023 - Dec 2024

- Led weekly in-person and online tutoring in calculus and linear algebra through a student-run mathematics program.

Ilearn Education

Senior Parliamentary and Lincoln-Douglas Debater

Irvine, CA

Aug 2019 - Dec 2023

- Built and delivered evidence-based arguments; led tournament preparation for 10+ peers and coached developing debaters.

Selected Projects and Honors

- **Desmos Bezier Renderer:** Built a real-time curve-rendering and animation workflow integrated with Desmos. Code
- **GitProfile Online Resume:** Built and deployed a responsive online resume highlighting selected research, technical work, and experience.
- **Illustrated Calculus Book:** Turned introductory calculus into a cartoon-based guide for middle and high school students and partnered with an art center for publication on Amazon.
- **Honors:** 1st Place Speaker, NSDA Stanford Invitational (LD); National Online Forensics Parliamentary Debate Champion; HiMCM Honorable Mention (top 28% of 1,055 teams); 2nd Place, USC SCMC Math Competition and CSULB Math Day at the Beach.

Skills

Programming: Python, Java, JavaScript, Lua, MATLAB, Git, LaTeX

Methods: NumPy, pandas, Matplotlib, regression, wavelet analysis, data visualization, research documentation

Mathematics: Differential equations, multivariable calculus, linear algebra, probability, statistics

Languages: English, Mandarin Chinese